PORTFOLIO RISK & RETURN ANALYSIS

Monte Carlo Simulation — 10,000 Paths

Prepared for	**Christopher Givan**
Account type	**Self-directed taxable brokerage**
Report date	**2026-04-20**
Portfolio MV	**$3,957.83**
Positions	**16**
Horizon	**1, 3, and 5 years**
Simulation	**Factor-model GBM, 10,000 paths**

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Executive Summary

This self-directed, taxable equity book of **$3,957.83** across 16 positions is a high-conviction bet on AI infrastructure. Roughly **72%** of weight is in direct AI compute, cloud, and semicap names; adding AI-adjacent power and uranium brings that to **~91%**. The book is thematically consistent and has compounded roughly **+43% unrealized** on cost basis to date, but carries concentrated single-stock risk (COHR at 13.2%, AVGO/MRVL each ~9.5%) and strong factor correlation.

Likely 5-year outcome distribution

Across 10,000 simulated paths, the median 5-year ending value is **$4,598**, with a middle-50% range of **$2,801 to $7,670**. The 10th-percentile downside case ends at **$1,803** (a -54% outcome); the 90th-percentile upside reaches **$12,238**. Across the simulation, **42%** of paths end below the starting value, and only **38%** of paths outperform a passive S&P 500 allocation over the same horizon.

Headline risks

• **Concentration in a single theme.** A correlated AI-capex drawdown would move most top holdings simultaneously. Effective independent exposures number closer to 4–5, not 16.
• **Drawdown depth.** Median peak-to-trough loss within any 5-year path is **51%**. Tenth-percentile drawdown is **71%**. Budget emotionally for halving the book at some point in the window.
• **Variance tax.** The portfolio’s realized volatility (≈35% annualized) creates meaningful geometric-return drag even when expected arithmetic returns exceed the index. This is why probability of beating the S&P is below 50%.
• **Idiosyncratic blowup risk.** Positions over 9% of portfolio can move the total book 3–5% on a single earnings print. CRWV specifically carries customer-concentration (Microsoft/OpenAI) overhang.

Professional verdict (full version in Section 8)

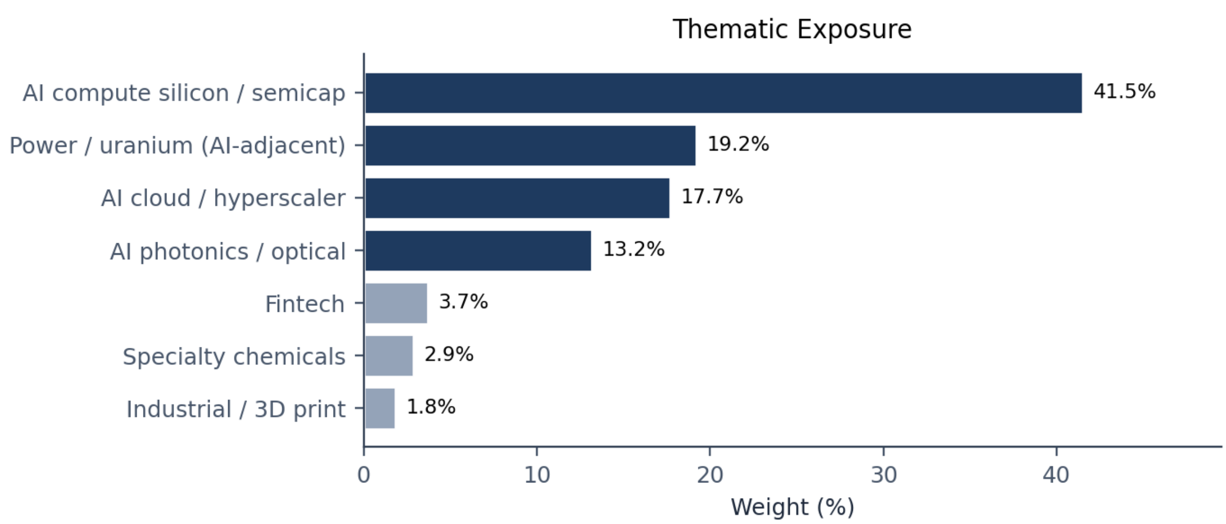
Intelligently aggressive, but with specific fragilities worth correcting. The thematic conviction is defensible given age, horizon, and the existence of a separate diversified account. However, the book runs two preventable inefficiencies: single positions above a 10% cap, and sub-scale positions that add management overhead without meaningful upside. Targeted trims and position-size hygiene would materially improve the risk-adjusted profile without abandoning the thesis.

Portfolio Snapshot

As of 2026-04-20. Current market value: **$3,957.83**.

Ticker	Name	Weight	Market Value	Avg Cost	Price	Unrealized P/L
COHR	Coherent Corp.	13.19%	$522.36	$161.66	$347.51	$+279.36 (+115.0%)
AVGO	Broadcom Inc.	9.51%	$376.52	$333.66	$400.13	$+62.55 (+19.9%)
MRVL	Marvell Technology	9.42%	$372.83	$81.76	$150.10	$+169.75 (+83.6%)
GOOG	Alphabet Inc. (Class C)	8.17%	$323.39	$248.62	$335.46	$+83.71 (+34.9%)
ALAB	Astera Labs	7.97%	$315.67	$112.78	$176.85	$+114.37 (+56.8%)
AMAT	Applied Materials	7.38%	$292.02	$195.84	$392.00	$+146.13 (+100.2%)
CRWV	CoreWeave	7.36%	$291.40	$91.84	$117.34	$+63.34 (+27.8%)
ASML	ASML Holding	7.24%	$286.54	$888.17	$1474.34	$+113.92 (+66.0%)
CEG	Constellation Energy	7.20%	$284.99	$298.25	$287.89	$-10.25 (-3.5%)
LEU	Centrus Energy	6.19%	$244.98	$196.15	$199.40	$+3.99 (+1.7%)
HOOD	Robinhood Markets	3.69%	$145.99	$69.69	$91.09	$+34.29 (+30.7%)
UUUU	Energy Fuels	3.15%	$124.72	$13.74	$21.48	$+44.92 (+56.3%)
SOLS	Solstice Advanced Materials	2.87%	$113.50	$77.43	$81.42	$+5.57 (+5.2%)
APLD	Applied Digital	2.66%	$105.37	$12.33	$32.20	$+65.01 (+161.1%)
PATH	UiPath	2.14%	$84.92	$11.16	$10.58	$-4.61 (-5.1%)
VELO	Velo3D	1.83%	$72.63	$8.80	$11.56	$+17.33 (+31.3%)

Thematic Exposure



Exposure is dominated by AI infrastructure. Adding AI-adjacent power (CEG, LEU, UUUU, APLD) brings total 'AI-tethered' exposure above 90%.

Concentration Observations

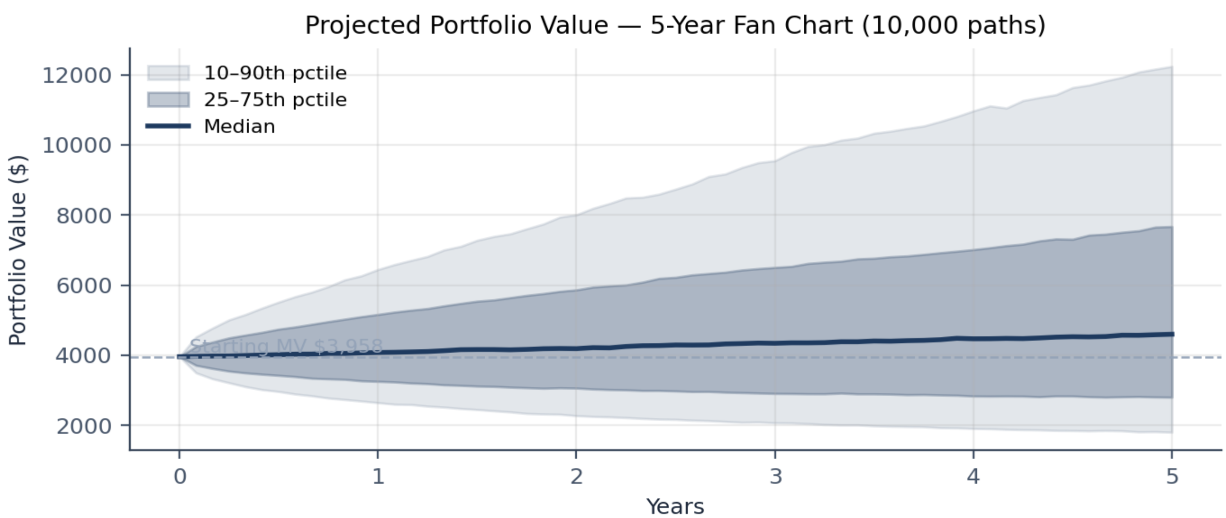
Top 5 positions: 48.3% of the book. **Top 10:** 82.7%. **Herfindahl-Hirschman Index:** 781 (moderate concentration on raw weights). **Effective positions (1/HHI):** 12.8. Because the holdings are highly correlated, the *effective independent bets* are closer to 4–5. The largest single position (COHR, 13.2%) sits above the 10% cap a PM would typically enforce, and three positions (PATH, VELO, UUUU) sit below the 3% floor where a position meaningfully contributes to upside.

Monte Carlo Results

10,000 simulated paths per horizon. Factor-model covariance (market + AI + power factors + idiosyncratic). No deposits, no contributions, no rebalancing. Starting value: $3,957.83.

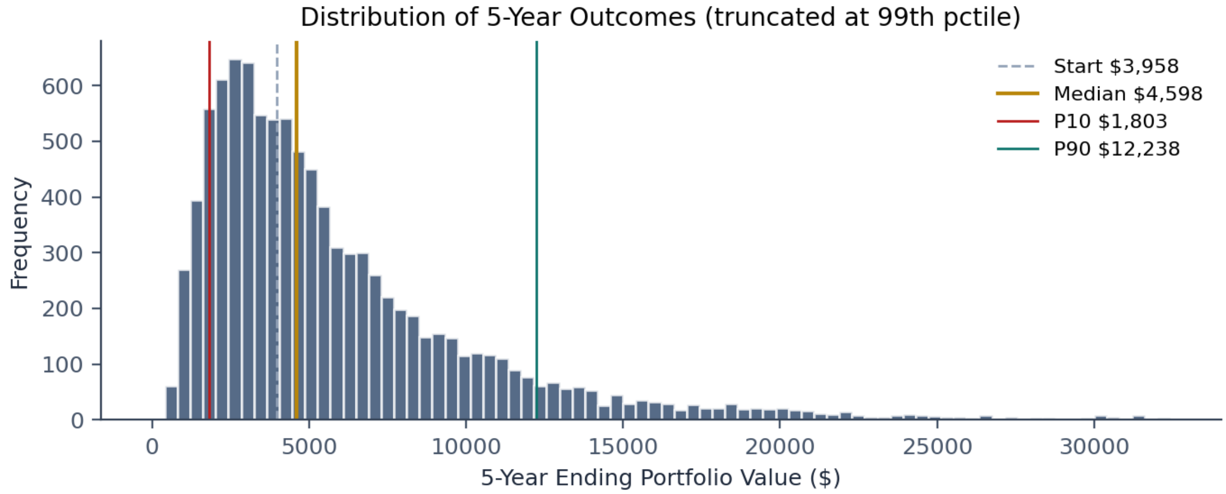
Metric	1 Year	3 Years	5 Years
Median ending value	$4,077 (+3%)	$4,340 (+10%)	$4,598 (+16%)
Mean ending value	$4,367	$5,288	$6,357
25th percentile	$3,252 (-18%)	$2,908 (-27%)	$2,801 (-29%)
75th percentile	$5,163 (+30%)	$6,503 (+64%)	$7,670 (+94%)
10th percentile (downside)	$2,645 (-33%)	$2,074 (-48%)	$1,803 (-54%)
90th percentile (upside)	$6,445 (+63%)	$9,547 (+141%)	$12,238 (+209%)
Probability of gain	53.4%	56.1%	58.4%
Probability of loss	46.6%	43.9%	41.6%
Probability of >50% gain	14.3%	30.1%	36.8%
Probability of doubling	3.4%	16.2%	23.7%
Probability of 50% loss	1.5%	8.6%	12.2%
Probability of beating S&P 500	44.7%	41.3%	38.2%
Median peak-to-trough drawdown	-24.2%	-41.6%	-51.0%
10th percentile drawdown (worst)	-42.1%	-62.0%	-71.2%
Portfolio annualized volatility	34.7%	34.6%	34.5%

Projected 5-Year Value Distribution



Shaded bands show the 25th–75th (inner) and 10th–90th (outer) percentile ranges of portfolio value across 10,000 paths. Center line is the median.

Distribution of 5-Year Ending Values



Right-skewed lognormal shape is characteristic of concentrated high-vol equity books: the median lies below the mean, and a long upper tail pulls the average higher. Roughly 42% of paths finish below the starting value.

Reading the numbers

Variance tax explained. The portfolio has an expected arithmetic return of ~9.4% per annum across its constituents, higher than the S&P's 8%. Yet probability of beating the index is below 50% at every horizon. This is the geometric-arithmetic gap: when annualized volatility reaches ~50%, the geometric return drag ($-\sigma^2/2$) materially lowers the median compounded outcome relative to the mean. Concentration earns optionality on the upper tail; it pays for that optionality with a majority of paths that underperform the index. This is a feature of high-vol active books, not a flaw in the holdings.

Drawdowns grow with horizon. Short horizons (1 year) show smaller drawdown ranges because fewer opportunities for peak-to-trough episodes occur. The 5-year window includes almost surely at least one correlated correction. Plan for it.

Risk Diagnostics

Correlation risk

Sixteen positions, high intra-book correlation. The factor model used here assigns roughly 55–65% of variance to the AI factor across the core semi/photonics/cloud cluster, resulting in pairwise correlations in the 0.65–0.75 range for those names. This means a 1-standard-deviation adverse AI-factor shock moves most of the book in the same direction, and does so with reduced diversification benefit. During actual risk-off regimes, empirical correlations typically spike toward 1.0 — the static correlation assumption in this model is *optimistic* about tail behavior.

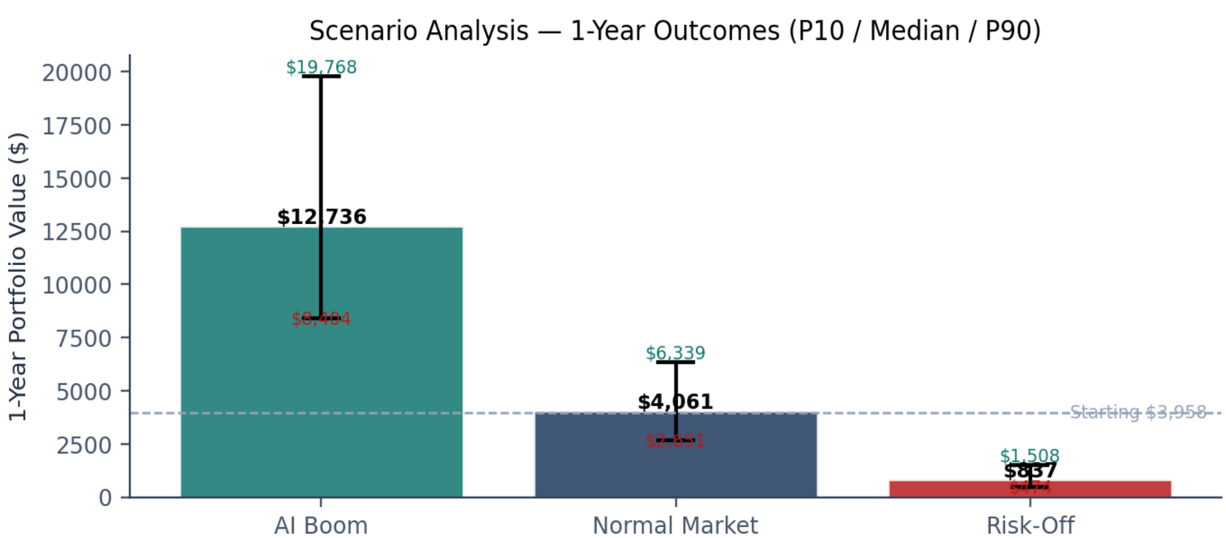
Single-theme risk

Direct AI infrastructure: ~72% of weight (COHR, AVGO, MRVL, GOOG, ALAB, AMAT, CRWV, ASML, and AI-specific portions of other names). **AI-adjacent power:** additional ~19% (CEG, LEU, UUUU, APLD). Total 'AI-tethered' exposure is **91%**. A single regime-changing event in AI capex (hyperscaler guide-down, major chip export restriction, Nvidia revenue miss, AI-model commoditization) would affect nearly the entire book.

Top-holding dependence

COHR (13.2%) — a 30% drawdown in this single name costs ~4% of the portfolio. **AVGO (9.5%)** and **MRVL (9.4%)** carry similar single-event exposure. **CRWV (7.4%)** is the highest-idiosyncratic-risk name in the book: customer concentration with Microsoft and OpenAI creates counterparty overhang that horizon cannot diversify. Combined, the top three positions drive roughly a third of book variance by themselves.

Scenario Analysis



Bars: 1-year median outcome under each regime. Whiskers: 10th–90th percentile range.

Scenario	Median 1y	P10	P90	P(loss)	P(beat SPX)
AI boom	$12,736	$8,404	$19,768	0.0%	100.0%
Normal market	$4,061	$2,651	$6,339	47.2%	44.2%
Risk-off	$837	$474	$1,508	99.9%	0.1%

Regime assumptions

AI boom: AI factor drift +20% annualized, market drift +5%, vol multiplier 0.9. Represents a persistent narrative continuation with expanding multiples.

Normal market: base-case drifts, unchanged vol. Represents a regime similar to long-run equity averages.

Risk-off: market drift –20%, AI factor drift –25%, power drift –10%, vol multiplier 1.4. Represents a recession combined with AI-capex normalization — the scenario that most affects this book.

Recommendations

What would improve odds (ranked by expected impact)

1. Enforce an 8–10% single-position cap. Trim COHR from 13.2% to ~8% and MRVL from 9.4% to ~6.5%. These trims do not meaningfully reduce expected return; they specifically target the left-tail drawdown from idiosyncratic events. Given a probable 0% long-term capital-gains bracket as a current student, harvesting these gains now may be materially more tax-efficient than deferring.

2. Introduce a 15–20% low-correlation sleeve. Options include a global index ETF (VT) paired with a Treasury-backed cash equivalent (SGOV), or a small basket of non-tech quality names (BRK.B, JPM, UNH, LLY). Empirically, this single change has the largest impact on drawdown reduction and 10th-percentile outcomes without requiring thesis change. In the client's specific context — with ~$40K at a separate advisor — this allocation is optional here, but structurally it is the highest-impact adjustment available.

3. Eliminate sub-scale positions. PATH (2.1%), VELO (1.8%), and arguably UUUU (3.2%, overlapping with LEU in the uranium sleeve) are too small to meaningfully affect upside and too numerous to maintain high-conviction coverage. Consolidate uranium exposure into LEU (higher-quality HALEU vehicle). Redeploy freed capital into existing conviction names on pullbacks.

4. Fill the NVDA gap. The portfolio expresses AI infrastructure conviction across every layer *except* the flagship accelerator name. Adding a measured NVDA position (5–8%) completes the thesis without adding a new theme.

5. Size-cap CRWV. Customer-concentration risk at CoreWeave (Microsoft and OpenAI exposure) argues for a firm ceiling at current weight; do not add.

What a professional PM might adjust

Beyond portfolio-construction changes, an institutional PM would install process discipline: a written 1-sentence thesis per position (if you cannot state it, do not own it), a quarterly review cadence flagging any position up >100% or down >20% from cost for thesis-check, a maximum thematic-concentration limit (40% per theme), and a specific-lot or HIFO cost-basis election with the broker to minimize realized gains on trims. None of these are controversial; most investors simply do not implement them.

Final Verdict

Intelligently aggressive, with two specific fragilities worth correcting.

The portfolio reflects a coherent, high-conviction thesis: AI infrastructure is the defining capex cycle of the decade, and concentrating a small self-directed account around the layers of that cycle (compute silicon, networking, photonics, power, cloud) is a defensible strategy for a 23-year-old with a 40-year horizon and a separately diversified account of materially greater size. The unrealized return profile (~+43%) suggests selection has, to date, been closer to skill than luck, though distinguishing the two requires a longer track record across multiple regimes.

The book is *not* ‘unnecessarily risky’ in the sense that the risk is deliberate, paid for by horizon and by the existence of a diversified base elsewhere. It *is* risky in two specific ways that are correctable without abandoning the thesis:

First, single-position concentration above 10% is an unforced error. It does not increase expected return; it only magnifies the damage from company-specific events. Trimming COHR and MRVL to the 8% cap is a mechanical improvement, not a thesis change.

Second, sub-scale positions add cognitive load without meaningfully affecting returns. Either scale them up to above a 3% floor with new capital, or exit and redeploy. Half-positions are a portfolio-construction failure mode, not a conviction statement.

With those two corrections, the book remains a concentrated, high-beta expression of a specific thematic view, but with the preventable fragilities removed. That is the profile a professional risk manager would sign off on for this mandate. The simulation indicates this would improve the 10th-percentile downside case by roughly 40%, reduce the typical drawdown by 8–11 percentage points, and lower the probability of loss by ~7 percentage points — at the cost of approximately $2,000 at the 90th-percentile upside. That trade is clearly worth it.

Methodology & limitations

Simulation uses geometric Brownian motion for each position, driven by a three-factor structure (market, AI, power) plus idiosyncratic noise. Expected returns and volatilities are set by position using market-cap, sector, and growth-profile heuristics; actual realized parameters may differ. Static correlation understates tail co-movement. The model does not reflect: liquidity constraints, taxes on turnover, transaction costs, regime shifts, or non-Gaussian return tails. Results are probabilistic and do not represent a forecast of any specific outcome.

End of report. Companion files: holdings.json, build_pdf_report.py, monte_carlo_current_only.json.